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Piero Lo Monaco, PhD
BMC Medical Research Methodology

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Dear Dr. Piero Lo Monaco,

We wish to submit an original research article entitled “Spatial Cluster Randomized Trials - Sampling Design with Spillover Effects & Spatial Dependence” for consideration by *BMC Medical Research Methodology*. We confirm that this work is original and has not been published elsewhere, nor is it currently under consideration for publication elsewhere. Our study addresses a critical gap in trial design by moving beyond post-hoc analysis to investigate the proactive role of treatment sampling design in managing spatial complexities.

In this paper, we report on a comprehensive simulation study that systematically compares the performance of simple random sampling (SRS) and block stratified sampling (BSS) methods. This research is significant because it provides evidence-based guidance for investigators on how to mitigate the confounding influences of spillover effects and spatial dependence from the outset of study planning. While existing literature focuses on post-hoc analytical adjustments, our work highlights how initial allocation strategies can be optimized to improve the validity of trial results.

Our findings indicate a fundamental trade-off between the two methodologies: while randomly selected treatment assignments can result in reduced Mean Squared Error (MSE) in best-case scenarios, block stratified assignments lead to minimal variation in MSE across different treatment combinations. This demonstrates that BSS is a robust strategy that ensures respectable estimation error remains consistent across a range of parameters, protecting against “worst-case” arrangements that could undermine statistical inference. We believe this manuscript is appropriate for publication by *BMC Medical Research Methodology* because it offers a novel comparative framework for designing more efficient and valid cluster randomized trials in complex spatial settings and it provides robust advice for study design to investigators developing cluster randomized trials with spatial considerations.

We have no conflicts of interest to disclose.

Please address all correspondence concerning this manuscript to me at awalther@unc.edu & my PhD advisor Dr. Feng-Chang Lin at flin33@email.unc.edu.

Thank you for your consideration of this manuscript.

Sincerely,

Andrew Walther